CSA1522 - CLOUD COMPUTING AND BIG DATA ANALYTICS
Experiment-4

4. Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

AIM:

To develop a simple Library Book Reservation System for the SIMATS library using Zoho Creator (Cloud Service Provider) to demonstrate the Software as a Service (SaaS) model. The application stores book details (title, author, publisher, year, edition, number of copies, rack number) and tracks book issue, return and reservation status.

PROCEDURE:

Log in to Zoho Creator.

Create a new application named Library Reservation Management System.

Create the form Book Registration with the following fields:

- Book Id, Book Title, Author Name
- Publisher Name, Publication Year, Edition, ISBN Number
- Book Category
- Total Copies, Available Copies
- Rack Number, Shelf Number
- Book Status (Available / Reserved / Issued)

Create the form Book Issue Return with the following fields:

- Student, Book, Book Id Ref
- Issue Date, Due Date, Actual Return Date
- Copies Issued, Book Condition
- Issue Status (Issued / Returned / Overdue), Fine Amount, Remarks

Create the form Book Reservation (Student, Book, Reservation Date, Required Date, Copies Requested, Reservation Status) and the form Student Details.

Create a workflow named Auto-Set Issue Status on Return on the Book Issue Return form (Edited, User input of Actual Return Date) with a Deluge script that sets the issue status to Issued or Returned.

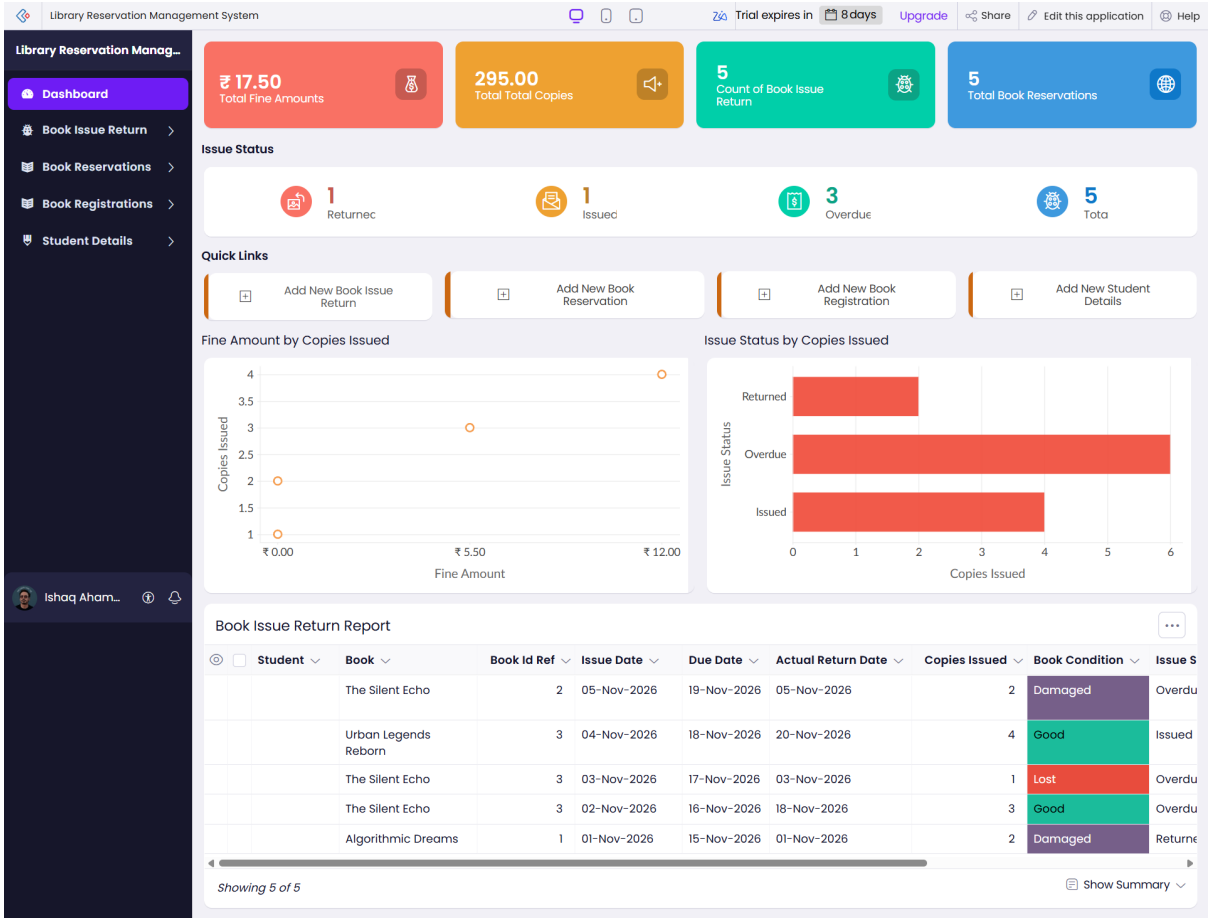
Create a Dashboard showing total fine amounts, total copies, issue status, charts and the Book Issue Return Report.

Save and publish the application.

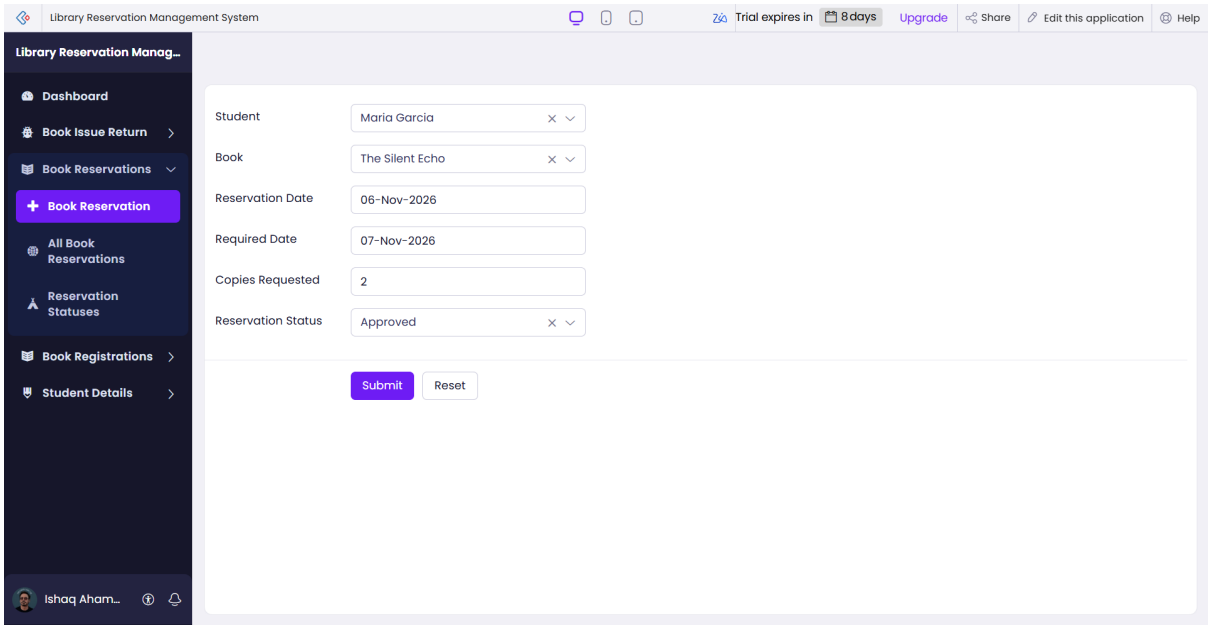
Enter sample book, issue and reservation details and submit the forms.

Verify the records in the reports and the dashboard.

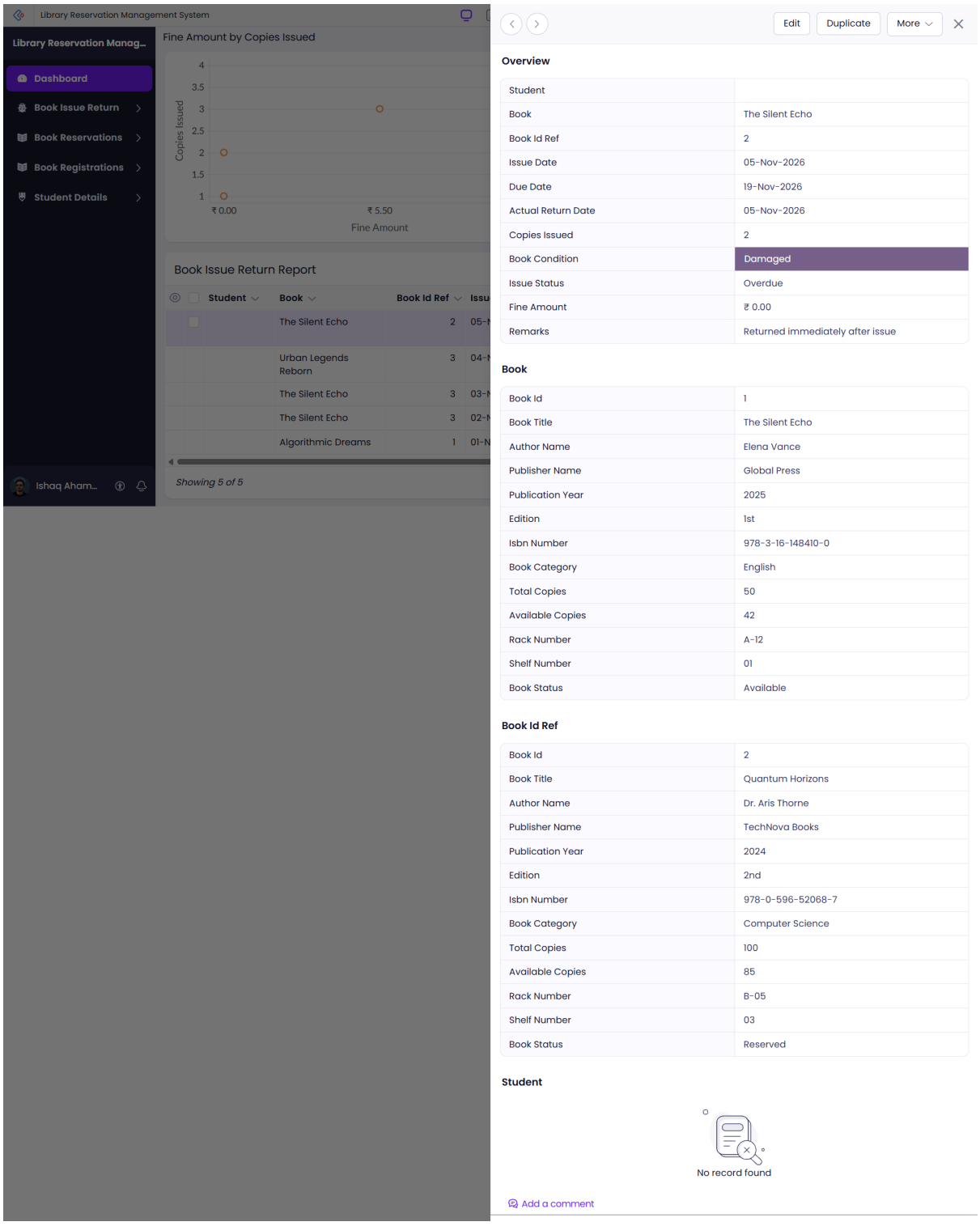
DESIGN:



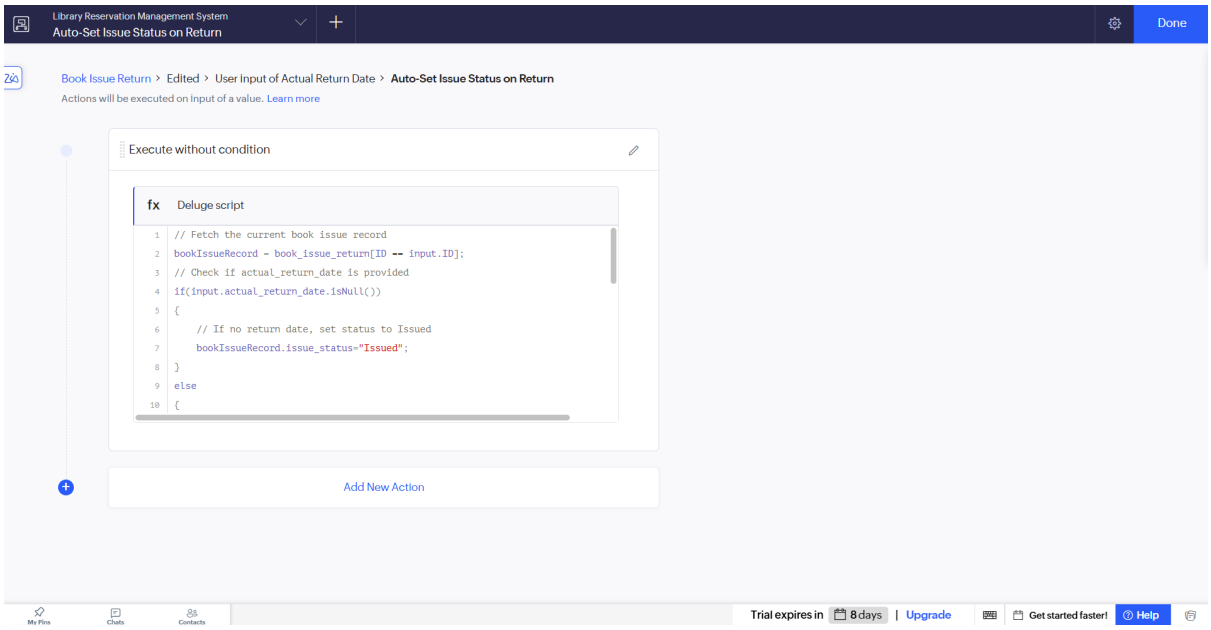
INPUT:



OUTPUT:


Library Reservation Management System
Library Reservation Manag...
Dashboard
Book Issue Return
Book Reservations
Book Registrations
Student Details
Ishaq Aham...
Fine Amount by Copies Issued
Copies Issued
Fine Amount
₹ 0.00
₹ 5.50
Book Issue Return Report
Student
Book
Book Id Ref
The Silent Echo
Urban Legends Reborn
Algorithmic Dreams
Showing 5 of 5
Edit
Duplicate
More
Overview
Student
Book
The Silent Echo
Book Id Ref
2
Issue Date
05-Nov-2026
Due Date
19-Nov-2026
Actual Return Date
05-Nov-2026
Copies Issued
2
Book Condition
Damaged
Issue Status
Overdue
Fine Amount
₹ 0.00
Remarks
Returned immediately after issue
Book
Book Id
1
Book Title
The Silent Echo
Author Name
Elena Vance
Publisher Name
Global Press
Publication Year
2025
Edition
1st
Isbn Number
978-3-16-148410-0
Book Category
English
Total Copies
50
Available Copies
42
Rack Number
A-12
Shelf Number
01
Book Status
Available
Book Id Ref
Book Id
2
Book Title
Quantum Horizons
Author Name
Dr. Aris Thorne
Publisher Name
TechNova Books
Publication Year
2024
Edition
2nd
Isbn Number
978-0-596-52068-7
Book Category
Computer Science
Total Copies
100
Available Copies
85
Rack Number
B-05
Shelf Number
03
Book Status
Reserved
Student
No record found
Add a comment

WORKFLOW:



RESULT:

A cloud-based Library Book Reservation System for the SIMATS library was successfully developed using Zoho Creator. The application stores book, student, issue and reservation details, automatically updates the issue status when a book is returned, and displays the records in reports and a dashboard. This demonstrates the Software as a Service (SaaS) model using a cloud platform.